

Automate
Dy-
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Anal-
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sis
Signature
Gen-
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Visual Roadmap

The Problem

Our Solution

Why
mal-
ware
sig-
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tures

mat-

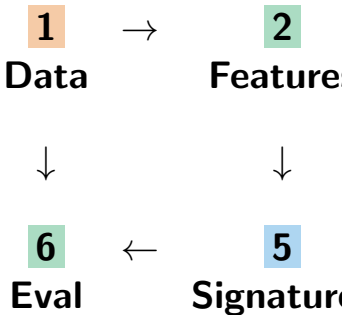
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The Man- ual Sig- na- ture Gen- er- a- tion Problem

Current
Goal

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The 6- Stage Pipeline



Stage 1 & 2: Data & Fea- ture Extraction

Feature
Dis-
trib-
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Stage

3:

Be-

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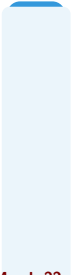
Representation

Key

In-

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Stage 4: LLM Se- man- tic Analysis



Using
Llama
(Open-
Source
Lo-
cal
LLM)

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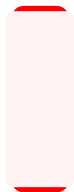
Automated Dynamic Analysis Why? Signature Generation

Stage 5 & 6: Sig- na- ture Gen- er- a- tion & Validation

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Real Chal- lenges We Faced



Challenge 1: JSON Hell

No

two

CAPEv2

JSON

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Signature Gen- er- a- tion & Qual- ity Metrics

Outputs

20
sig-
na-

Cross-Family Generalization

Tested

(New

Fam-

Trained

(Known

Fam-

.

77%

Pre-

86%

Pre-

65%

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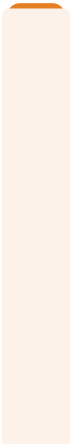
MITRE ATT&CK Tech- nique Mapping

Coverage
by
Tech-
nique

Pro-
cess
In-
jec-
tion:

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Real-World Impact



**Security
Op-
er-
a-
tions
Cen-
ters
(SOCs)**

From
4+
hours
per

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Key Takeaways

1

Feasible

—

LLMs

can

gen-

er-

ate

production-

quality

mal-

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Lessons Learned

What
To

Im-
prove
What
Worked



Prototype
Aggressive
with
caching
smaller
(saved
dataset
2
first
days)



Document
Modular
data-
note-
for-
book

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Honest Limitations

What
This
Sys-
tem
CAN
Do:



Generate
use-
ful
sig-
na-
tures

Repository:

[https://github.com/ykram051/](https://github.com/ykram051/Analysis-Signature-Generatio)
[Analysis-Signature-Generatio](https://github.com/ykram051/Analysis-Signature-Generatio)

*All
code,
sig-
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tures,
and
eval-
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tion
re-
sults
avail-
able
on*

GitHub

